PRX-Foil Series — Ultra-Precision Bulk Metal Foil Resistors

Synthetic test datasheet for KB ingestion pipeline validation

Parameter	Value	Condition
Resistance Tolerance	0.01% (100 ppm)	standard grade
Temperature Coefficient (TCR)	0.2 ppm/C typical	-55C to 125C
Long Term Stability	25 ppm / year typical	rated power, 25C
Voltage Coefficient	< 0.1 ppm/V	negligible nonlinearity
Thermal EMF	< 0.05 uV/C	low thermoelectric error
Power Rating	0.3W to 0.6W	package dependent
Current Noise Index	< -40 dB	extremely low excess noise

1. General Description

The PRX-Foil series are bulk metal foil resistors designed for applications demanding the highest levels of precision and long term stability. The foil resistive element exhibits near zero temperature coefficient of resistance and extremely low current noise compared to thin film or thick film resistor technologies, making this series the preferred choice for precision current sense applications, voltage references, and gain-setting networks in low noise instrumentation.

2. Why Resistor Selection Matters for Current Sources

In a precision current source, the sense or gain resistor directly sets output current accuracy and contributes Johnson thermal noise to the total output noise budget. A resistor with high temperature coefficient will cause current drift over temperature even if the amplifier and reference are ideal. Selecting an ultra-precision foil resistor with sub 1 ppm/C TCR for the critical gain-setting resistors in a Howland or Libbrecht-Hall style current pump significantly improves both the absolute accuracy and the temperature stability of the generated current.

3. Matching and Tracking

For topologies that rely on resistor ratio matching, such as the Howland current pump, using resistors from the same production batch or a matched resistor network array improves ratio tracking far beyond what individual tolerance specifications alone would suggest, because correlated drift between matched elements cancels in the ratio.